for engulfing pathogens by pseudopodia formed by the streaming movement of the cytoplasm.

Ciliary movement - This type of movement occurs in the respiratory passages and genital tracts which are lined by ciliated epithelial cells.

Flagellar movement - This type of movement occurs in the cells which are having flagella or whip-like motile organelle. The sperm cells show flagellar movement.

Muscular movement -The movement of hands, legs, jaws, tongue are caused by the contraction and relaxation of the muscle which is termed as the muscular movement.

9.2. Types of muscles

Muscles are specialized tissues which are derived from the embryonic **mesoderm**. They are made of cells called **myocytes** and constitute 40 – 50 percent of body weight in an adult. These cells are bound together by a connective tissue to form a muscular tissue. The muscles are classified into three types, namely **skeletal, visceral** and **cardiac muscles.**

9.3 Skeletal muscle (Voluntary muscle)

Skeletal muscle is attached to the bone by a bundle of collagen fibres known as **tendon**. Each muscle is made up of bundles of **muscle fibres** called **fascicle**. Each muscle fibre contains hundreds to thousands of rod-like structures called **myofibrils** that run parallel to its length. The connective tissue covering the whole muscle is the **epimysium**, the covering around each fascicle is the **perimysium** and the muscle fibre is surrounded by the **endomysium**. They control the voluntary actions such as walking, running, swimming, writing hence termed as voluntary muscles.

9.3.1. Structure of a skeletal muscle fibre

Each muscle fibre is thin and elongated. Most of them taper at one or both ends. Muscle fibre has multiple oval nuclei just beneath its **plasma membrane** or **sarcolemma**. The cytoplasm of the muscle fibre is called the **sarcoplasm**. It contains glycosomes, myoglobin and sarcoplasmic reticulum. **Myoglobin** is a red- coloured respiratory pigment of the muscle fibre. It is similar to haemoglobin and contains iron group that has affinity towards oxygen and serves as the reservoir of oxygen. **Glycosomes** are the granules of stored glycogen that provide glucose during the period of muscle fibre activity. Actin and myosin are muscle proteins present in the muscle fibre.

Along the length of each myofibril there are a repeated series of dark and light bands (Figure 9.1). The dark **A-bands** (Anisotropic bands) and the light **I-bands** (Isotropic bands) are perfectly aligned with one another. This type of arrangement gives the cell a striated appearance. Each dark band has a lighter region in its middle called the **H-Zone** (H-Helles: means clear). Each H-zone is bisected vertically by a dark line called the M-line (M-for middle). The light I-bands also have a darker mid line area called the **Z–disc** (from the German

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